Q-HEALTH GUARD

Privacy-Preserving Federated Quantum Healthcare AI

Project Documentation | Problem Statement VNQFF-02 | Team: Epic Engineers

1. Project Overview

Q-HEALTH GUARD is a proposed healthcare AI research workspace exploring how multiple hospitals could collaborate on prediction models without directly sharing raw patient records. The design combines local data preparation, experimental Quantum Machine Learning (QML), Federated Learning, privacy-aware model-update handling, and a dashboard for viewing experiment results.

Current status: the existing project material describes a frontend/demo interface. Real model execution, federated training, and production-grade privacy/security controls must be implemented and tested before they can be described as working features.

2. Problem Statement

Hospitals hold sensitive patient information, such as medical histories, test results, and disease records. Sharing identifiable records between institutions can create privacy, security, ethical, and governance risks. Because each hospital manages its own data, models trained independently may not benefit from learning across different hospital datasets. Even when raw records are not exchanged, model updates may still reveal information if they are not appropriately protected.

> **Main problem to solve: How can multiple hospitals collaboratively train and evaluate a useful healthcare AI model without directly sharing raw patient records, while reducing the risk that exchanged model updates reveal sensitive information?**

3. Proposed Solution

Q-HEALTH GUARD proposes that each hospital keep its patient data locally and train a local model. Where suitable, the model can use a quantum kernel or parameterized quantum circuit and should be compared with a classical baseline. Participating clients can share permitted model updates rather than raw records; a central aggregator combines compatible updates and returns a global model. Update clipping and calibrated Differential Privacy can be evaluated to reduce leakage risk.

Important distinction: Federated Learning does not automatically guarantee privacy. Differential Privacy, secure aggregation, and encrypted communication are separate measures that must be correctly configured, implemented, and tested.

4. End-to-End Workflow

The diagram shows the proposed flow from local hospital data to dashboard results. It is an architecture plan; it does not prove that every stage is already implemented.

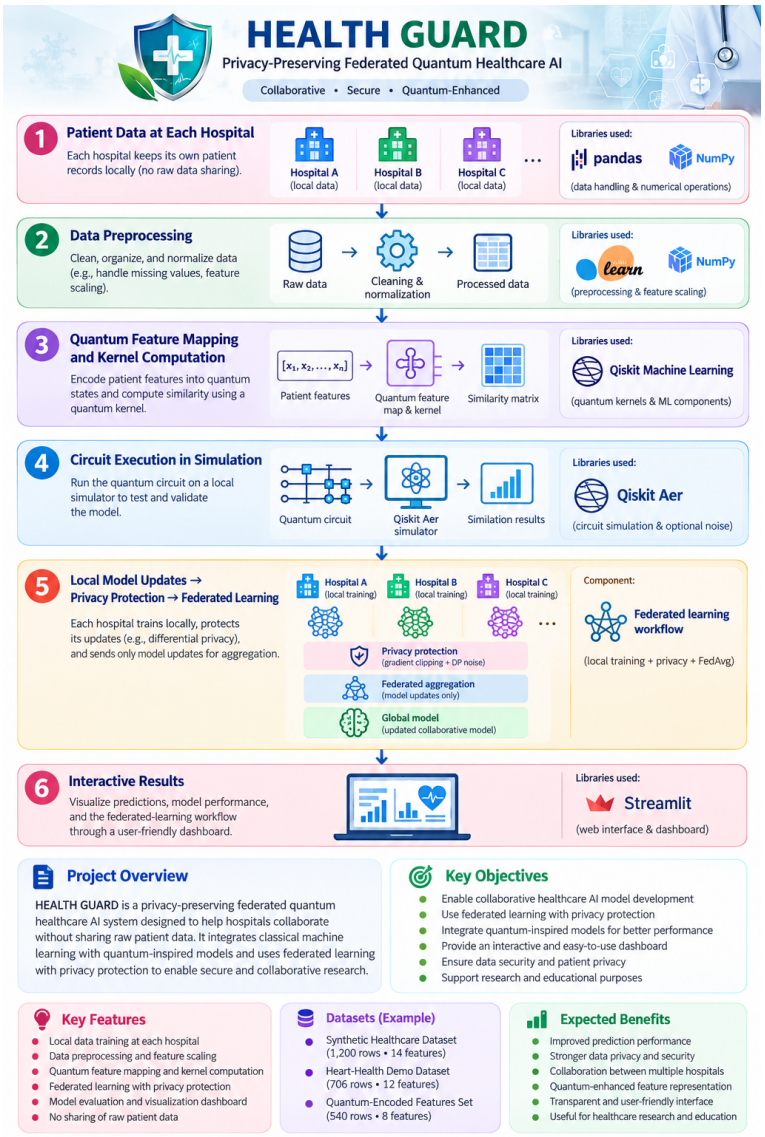


Figure 1. Proposed Q-HEALTH GUARD workflow.

Step 1 — Patient data stays local

Each hospital keeps raw patient records on its own system. Use synthetic or properly de-identified data in the prototype and public demonstrations.

Step 2 — Data preprocessing

Validate columns, handle missing values, encode categorical fields if needed, and scale numeric features. Fit preprocessing on training data only to avoid data leakage.

Step 3 — Quantum feature mapping and kernel

Select relevant features and encode them with a Qiskit quantum feature map. A quantum kernel can calculate sample similarities for a QSVC-style classification experiment.

Step 4 — Circuit simulation

Run quantum circuits with Qiskit Aer during development. Record dataset source, configuration, random seed, selected features, and runtime for reproducibility.

Step 5 — Privacy-aware model collaboration

For compatible trainable models, clients train locally, protect updates with clipping and appropriately calibrated Differential Privacy, and aggregate updates with a method such as FedAvg. A kernel-based QSVC workflow may need a different collaboration strategy; FedAvg does not automatically fit every model type.

Step 6 — Dashboard results

Display calculated predictions, evaluation metrics, charts, and experiment history after connecting a real backend. Mark placeholder values clearly as illustrative.

5. Main Components and Technologies

The following table lists the role of each suggested component. The table describes the intended stack, not necessarily features already connected in the current interface.

Component	Technology / method	Purpose
Data handling	Pandas, NumPy	Load, inspect, and prepare tabular data.
Preprocessing and baseline	Scikit-learn	Feature scaling, classical SVM baseline, and evaluation.
Quantum ML	Qiskit Machine Learning	Experiment with quantum kernels and QSVC-style classification.
Circuit simulation	Qiskit Aer	Run quantum circuits in a local simulator.
Federated workflow	Multiple simulated clients; FedAvg where compatible	Combine supported model updates without centralizing raw records.
Privacy protection	Update clipping, Differential Privacy	Reduce the risk of information leakage; parameters must be calibrated.
Dashboard	Streamlit	Present experiments, metrics, and visualizations.

6. Model Evaluation

Compare the quantum-kernel model with a classical baseline using a reproducible train/test or validation pipeline. Suitable metrics include accuracy, precision, recall, F1-score, and ROC-AUC when appropriate. Record class balance, dataset provenance, random seed, preprocessing steps, and runtime. Do not present illustrative dashboard numbers as measured results.

7. Privacy and Security

- Use synthetic or properly de-identified data during development.
- Never commit identifiable patient records or direct identifiers to a public repository.
- Federated Learning reduces the need to centralize raw data, but does not itself guarantee privacy.
- Clipping alone is not Differential Privacy; use a defined mechanism and document its parameters.
- Production systems need encryption, authentication, access controls, key management, audit logging, and retention policies.
- Quantum computing is an experimental modelling option—not a privacy guarantee or proof of better accuracy.

8. Implementation Plan

Recommended order of development:

- Implement dataset loading, preprocessing, and a classical SVM baseline.
- Build a Qiskit quantum-kernel experiment and compare it fairly with the baseline.
- Connect the dashboard to actual model execution and calculated metrics.
- Simulate multiple hospital clients and choose a federated method compatible with the selected model.
- Add and test privacy mechanisms and secure communications; measure the privacy–utility trade-off.
- Document experimental results, limitations, and reproducible setup instructions.

9. Current Limitations and Disclaimer

The supplied project material describes a frontend/demo interface; it does not confirm that backend model execution, secure database integration, real federated training, or production-grade privacy controls are complete. These capabilities require implementation and validation. HEALTH GUARD is for research and education only and must not be used to diagnose, treat, or make clinical decisions about patients.

10. Team Details

Team Name	Epic Engineers
Team Members	1. Jovial Korivi 2. Jhansi Potipadu 3. Lakshmi Manaswi Jampani 4. Jeswanth Aradhyula
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